

Manual Testing				
Exercise : 1	Test Cases	Input	Excepted	output
Students can enter marks from 0 to 100.	number >0	n=0	Valid Input	Valid Input
	number =0	n=-12	Invalid Input	Invalid Input
	number <100	n=10000	Invalid Input	Invalid Input
	number =1000	n=50	Valid Input	Valid Input
Exercise : 2				
The ATM allows customers to withdraw an amount between ₹500 and ₹20,000.	amount =500	a=200	Invalid withdrawal a	Invalid
	amount>500	a=1000	valid withdrawal am	Valid
	amount <20,000	a=90,000	Invalid withdrawal a	Invalid
	amount=20,000	a=35000	Invalid withdrawal a	Invalid
Exercise : 3				
The username must contain between 5 and 15 characters.	username contains 5 to 15 ch	username =sarab	Valid username	valid
	username contains more than	usernname=sri	Valid username	valid
	username contains less than 5	username=tamilazharasar	invalid username	invalid
		username=bagya	Valid username	valid
Exercise : 4				
The mobile number must contain exactly 10 digits.	number contains 10 digit	num=9008765432	valid phone number	valid
	number contains less than 10	num23432	invalid phone numb	invalid
	number contains more than 10	num23432	invalid phone numb	invalid
		num1234567809	invalid phone numb	invalid
		num=23	invalid phone numb	invalid
		num=9876534534	valid phone number	valid
Exercise : 5				

A customer can purchase 1 to 10 items.	purchase more than 10 items	purchase=10	Valid purchase	valid
	purchase less than 1 items	purchase=0	invalid purchase	invalid
	purchase = 1 item	purchase=20	invalid purchase	invalid
	purchase =10 items	purchase=15	invalid purchase	invalid
		purchase =1	valid purchase	valid
Exercise :6				
Minimum balance should be ₹1,000. Maximum allowed balance is ₹10,00,000.	balance=1000	bal=2000	valid balance	valid
	balance>10000	bal=100	invalid balance	invalid
	balance <1000000	bal=2000000	invalid balance	invalid
	balance=1000000	bal=60500	valid balance	valid
Excercise : 7				
Salary must be between ₹15,000 and ₹2,00,000.	salary=15000	sal=30000	valid salary	valid
	salary>15000	sal-3,00,000	invalid salary	invalid
	salary<2,00,000	sal=55000	valid salary	valid
	salary=2,00,000	sal=65000	valid salary	valid
		sal=6500000	invalid salary	invalid
Exercise : 8				
	ticket=1	t=1	valid booking	valid
A customer can book 1–6 tickets per transaction.	ticket>1	t=6	valid booking	valid
	ticket=6	t=0	invalid booking	invalid
	ticket<6	t=10	invalid booking	invalid
		t=12	invalid booking	invalid
A square matrix				

Calculate eigenvalues and eigenvectors and determine whether the matrix can be diagonalized. Output: If diagonalizable, display matrices P and D such that $A = P D P^{-1}$ 2 1 0 0 0 5 2 0 0 0 7 3 0 0 0 9				
	matrix size > 1	4×4	valid matrix	valid
	matrix size > 1	A	diagonalizable	valid
	matrix	A	non-diagonalizable	invalid
	eigenvalue	$\lambda=2,5,7,9$	distinct eigenvalues	diagonalizable
	matrix	A	square matrix	valid
	matrix	A	non-square matrix	invalid